



Maritime &
Coastguard
Agency

Guidance

Safety Bulletin 21 – Use of emery cloth on metalworking lathes

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This publication is available at <https://www.gov.uk/government/publications/safety-bulletin-21-use-of-emery-cloth-on-metalworking-lathes/safety-bulletin-21-use-of-emery-cloth-on-metalworking-lathes>

Following an incident involving the use of a lathe machine on a bulk carrier, which resulted in a broken arm, the MCA is issuing this safety bulletin to raise awareness of the risks of using emery cloths. The incident occurred when the victim placed a strip of emery cloth over the work piece of the metalworking lathe, and their boiler suit sleeve became caught in the fast-spinning shaft. The use of emery cloths on metalworking lathes is not recommended, and where it must be used, precautions should be taken.

Summary of issue

When using lathes, operators are exposed to moving machine parts that can crush and dismember, cause dislocations and broken bones, or even cause death. Accidents occur when hand-held emery cloth is entangled in the rotating parts of the machine or where the emery cloth is snagged on the rotating component dragging the operator into the danger zone.

Risk assessment should form the basis of all safety measures, and it is important to assess the risk of using emery cloth to polish, deburr or size a metal component while it is rotating on a manual lathe. Health and Safety Executive guidance states that emery cloth should never be applied directly by hand on manual metalworking lathes. If after risk assessment, it is not possible to eliminate the use of emery cloth completely, you should only apply it using a tool post or tool post and holding device combined.

The recent incident onboard a bulk carrier indicates that not everyone operating lathes onboard ships are aware of the risks and necessary precautions when using this machine. It highlighted the need for those working with lathes to be made aware of the risks and using emery cloths.

Safety Management Systems need to provide guidance on the safe use of workshop machinery, with specific controls for the lathe, which may include:

- prohibition of emery cloths, where reasonably practicable

- prohibition of the application of emery cloth by hand on a lathe machine, and precautions to take when using tools to apply emery cloth
- for computer numerically controlled (CNC) lathes, prohibition of application of emery cloth by hand or with a tool
- control movement of other personnel around the lathe while in operation
- avoiding adjustment of lathe, tool rest or setup while lathe is running
- requirement for those using the lathe to have adequate training and awareness of best practices

In addition, the following precautions should be noted to avoid entanglement in the rotating parts of machinery:

- You should not use gloves whilst applying emery cloth
- You should wear tight fitting overalls (preferably with elasticated cuffs) and avoid loose, baggy clothing and ties
- You should tie back long hair or wear an appropriate hair net
- You should remove rings, jewellery etc.

Actions to take

Shipowners and workers using lathes should be aware of best practice, including avoiding the use of emery cloth where practicable. Following risk assessment, precautions should be taken and laid down in operating procedures. All workers that may use a lathe should be aware of the risks, the precautions to take and be appropriately trained.

Legal requirements

The shipowner and employer must have regard to the provisions of the Merchant Shipping and Fishing Vessels (Health and Safety at Work) Regulations 1997 which set out the general requirements for health and safety at work.

All tools, machinery and equipment used at work ('work equipment') are covered by the requirements of The Merchant Shipping and Fishing Vessels (Provision and Use of Work Equipment) Regulations 2006 (PUWER).

Further information

The use of emery cloth on metalworking lathes, HSE information sheet, Health and Safety Executive is available on the HSE website at the bottom of The use of emery cloth on metalworking lathes (<https://www.hse.gov.uk/engineering/lathes.htm>) page.

Related Documents

MGN 331 (M+F) Amendment 1 provision and use of work equipment regulations 2006 (<https://www.gov.uk/government/publications/mgn-331-mf-amendment-1-provision-and-use-of-work-equipment-regulations-2006>)

MGN 636 (M) Amendment 1 merchant shipping and fishing vessels (health and safety at work) regulations 1997 (<https://www.gov.uk/government/publications/mgn-636-m-amendment-1-merchant-shipping-and-fishing-vessels-health-and-safety-at-work-regulations-1997>)



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